Econometric 2026, GE08: Assignment 1

5-2-2026 Submission Date:19-2-2026

Estimation of the Energy (or Calories), Protein, and Fat Demand Elasticity for Rural Areas. The data set is based on the 2023-24 Household Consumer Expenditure Survey (HCES-2) and the broad patterns for the nutritional intake is available in the NSSO Report No 594.

Data: The relevant variables are in the file asgn1-cal-prot.dta in the Asgn 1

State: Each team has to work on the state assigned to that team in Teams sheet

Software: The statistical software for the data analysis can be STATA, R or Python

Results format: The results have to summarised using the asdoc command (code will be illustrated in the class) for those using STATA. Other can use appropriate command to present the results.

The result include intercept, its standard error, slope and its standard error along with the * to indicate significance at the appropriate level of significance. Include R^2 and adjusted R^2 (wherever relevant) as well.

(1) Estimate the protein demand model using the following dependent (first variable) and independent variables (the remaining variables) using sampling weights (per_wgt= hh_size*weights). Summarise your findings using R^2 or Adjusted R^2 (wherever applicable), and the interpretation of the slope coefficient based on marginal effect and elasticity.

(a) ProteinHH (Y), mpce (X)
(b) Per capita ProteinHH (Y), mpce (X)
(c) ln(Per capita ProteinHH) (Y), ln_mpce (X)
(d) ln(Per capita ProteinHH) (Y), hh_size (X)
(e) ln(Per capita ProteinHH) (Y), hh_size(X2), ln_mpce (X3)

(2) (a) Estimate the following two additional models for energy and fat as in (e) above. Report these results for Energy, Protein and Fat together in one table and compare the elasticities.

ln(Per capita EnergyHH) (Y), hh_size (X2), ln_mpce (X3)
ln(Per capita FatHH) (Y), hh_size (X2), ln_mpce (X3)

(b) Test if the each of the energy, protein and fat demand is inferior or normal or luxury 'good'?

(3) Test which of the following two models fit the data better and interpret the coefficient of household size as in (a) and (b).

(a) ln(Per capita ProteinHH) (Y), hh_size (X21), ln_mpce (X3)
(b) ln(Per capita ProteinHH) (Y), ln(hh_size) (X22), ln_mpce (X3)

If the model fit is very similar for either of the models which one would you prefer to use for ease of interpretation and why?

(4) In the regression model of ln(Per capita ProteinHH) regressed on hh_size and ln_mpce, compare the results for the model with and without intercept and comment.

Submission Format

(1) The answers to the questions below has to be submitted as a pdf file. The name of the file should be T#_statename.pdf where the state name refers assigned state. For example, the first team will have the file name as T1_ANP.pdf if the state assigned is ANP (Andhra Pradesh).

(2) The first page should include the names of the team members and the name of the state. In the end of the document, add a section with title Appendix where the Python/R/STATA codes should be copied question-wise.

(3) You will receive a google form link closer to the deadline to submit the pdf file of this assignment. In the google form there will also be a link to submit the do file for STATA or the R codes as text file or the Python codes. Name the file here also as T#_statename.ext with 'ext' as per the software's extension for file name (e.g. ext =do for the STATA do file).

(4) The results have to be populated in the Compiled results.xlsx for each state as per the column headings and for one case it is illustrated.